* Hamming Distance :

- The hamming distance between two words (of same size) is the number of differences between the corresponding bits.
- We show the hamming distance between two words x and y as $d(x,y)$.

The hamming distance can easily be found if we apply the XOR operation ($\oplus$) on the two words and count the number of 1s in the result.

* Minimum Hamming distance

- The minimum hamming distance is the smallest hamming distance between all possible pairs.
- denoted by d_{min}.

Q Find d_{min} for 00000, 01011, 10101, 11110.

$d(00000, 01011) = 3$
$d(00000, 10101) = 3$
$d(00000, 11110) = 4$
$d(01011, 10101) = 4$
$d(01011, 11110) = 3$
$d(10101, 11110) = 3$.

$d_{min} = 3$

* If we want to detect upto S bit errors between send and received code word we need to distance between valid codewords to be S+1.

i.e if d_{min} between valid codewords is $S+1$, the received codeword cannot be mistaken for anothe codeword.

* When a received codeword is not a valid codeword, the receiver needs to decide which valid codeword was sent.
- The decision is taken based on concept of territory, Each valid codeword has its own territory, we use geometric approach to define its territore.

- To guarantee correction of upto t errors the minimun hamming distance must be

$$d_{min} = 2t+1$$

i.e (3, 5, 7 ...)

if $d_{min} = 4$

Correction bit = 1 bit
Identificat unto = 3 bits.

* Parity Bit - Error detection technique.

A parity bit is a single bit added to a group of data bits to make the total number of 1's either

- Even → Even Parity
- Odd → Odd Parity

- In this a k-bit dataword is changed to k+1 bit codeword where the extra bit is called parity bit.

- The minimum hamming distance for this category is $d_{min} = 2$, which means that the code is a single-bit error detecting code.
- It cannot correct any error.

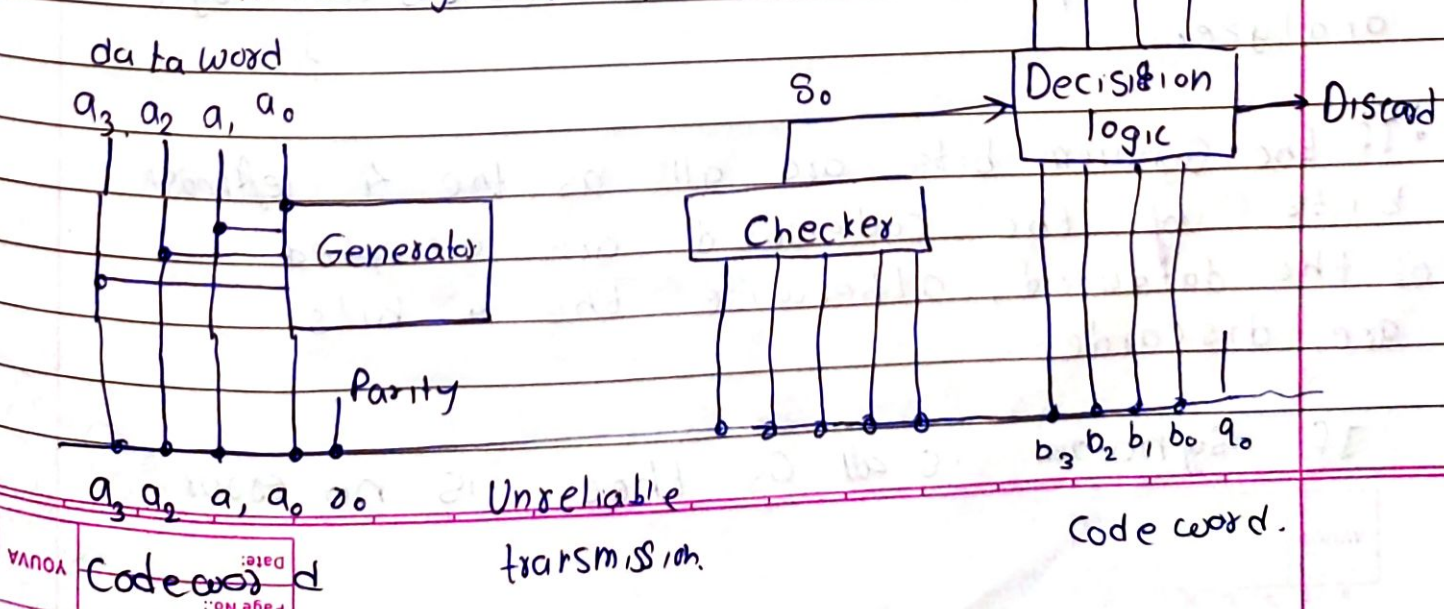
Data Word	Code word
0000	00000
0001	00011
0010	00101
0011	00110
0100	01001
0101	01010.

$r_0 = a_3 + a_2 + a_1 + a_2 \ \% \ 2$ (at sender)

If the number of 1s is even, the result is 0, if the number of 1s is odd, the result is 1.

$S_0 = b_3 + b_2 + b_1 + b_0 + q_0 \ \% \ 2$ (at receiver)

S_0 is called Syndrome.



* A Simple parity check can detect an odd number of errors.

* Cyclic Redundancy Check:

- In encoder, the datword has k bits, the Code word has n bits.
- The size of the dataword is augmented by adding n-k 0s to right hand side of the word.
- The resulting n bit word is fed to the generator.
- The generator divides the augmented dataword by the divisor.
- The quotient of the division is discarded, the remainder is appended to the dataword to create code word.

- The decoder receives the possible corrupted codeword. A copy of all n bits is fed to the checker which is a replica of the generator. The remainder produced by the checke is a syndrom of n-k bits which is passed to the decision logic analyzer.

- If the syndrom bits are all 0s, the 4 leftmost bits of the codeword are accepted as the dataword, otherwise the 4 bits are discarded.

If syndrom is all 0s, there is no error.

ex. data word - 1001

Encoder:

```
              1 0 1 0
1011 | 1 0 0 1 0 0 0
       1 0 1 1
       0 0 1 0 0
           0 0 0 0
           1 0 0 0
           1 0 1 1
           0 0 1 1 0
             0 0 0 0
             0 1 1 0
```

Code word : 1001110

↳ 1000110 → error.

Decoder-

```
              1 0 1 1
1011 | 1 0 0 0 1 1 0
       1 0 1 1
       0 0 1 1 1
           1 0 1 1
           0 1 0 0 0
             1 0 1 1
             0 0 1 1   → discard
```

```
              1 0
1011 | 1 0 0 1 1 1 0
       1 0 1 1
       0 0 1 0 1 1
           1 0 1 1
           0 0 0 0     → Accept
```

* Check Sum:

Checksum is a simple error detection technique where

- data is divided into equal ~~detection~~ sized blocks.
- All blocks are added together (using 1's complement arithmetic)
- The Ones complement of Sum becomes the checksum.

- At receiver, all blocks + Checksum are added again and Ones complement is taken
- If result is all zeros, then no error.

```
  1 0 1 1 0 0 1 1
  0 1 0 1 1 0 1 0
  0 1 1 1 0 0 1 0
  1 0 1 1 1 1 0 0
  0 0 0 1 0 1 0 1
  0 0 1 1 0 1 1 0
  ---------------
  1 0 0 0 0 1 1 0
          1 1 1 1
                1
  ---------------
  1 0 0 0 1 0 0 0
```

1's Comp : $\boxed{0111\ 0111}$

1 1

1 0 1 1 1 1 1

1 0 1 1 0 0 1 1
0 1 0 1 1 0 1 0
0 1 1 1 0 0 1 0
1 0 1 1 1 1 0 0
0 0 0 1 0 1 0 1
0 0 1 1 0 1 1 0
0 1 1 1 0 1 1 1

1 1 1 1 1 1 0 1
1 0

1 1 1 1 1 1 1 1

1's Comp : 0 0 0 0 0 0 0 0 ✓

Accepted

* Standard Ethernet :

- The original Ethernet was created in 1976 at Xerox's PARC. Since then, it has gone through four generations.
  - Standard Ethernet (10 Mbps)
  - Fast Ethernet. (100 Mbps)
  - Gigabit Ethernet (1 Gbps)
  - Ten Gigabit Ethernet. (10 Gbps)

* MAC Sublayer.

- In Standard ethernet, the Mac Sublayer governs the operation of the access method.
- It also frames data received from the upper layer and passes them to the physical layer.

* Frame Format

- The ethernet frame contains seven fields, preamble, SFD, DA, SA, length or type of protocol data unit (PDU), upper-layer-data, and CRC.

- The Ethernet does not provide any mechanism for acknowledg~~ments~~ing received frames, making it unreliable medium

7 bytes	1 byte	6 byte	6 byte	2 byte	min 46 bytes max 1500 bytes	4 bytes
Preamble	SFD	Dest Add	Source Add	Length or type	Data & padding	CRC
					Payload	

fig. 802.3 MAC frame.

"Standard Ethernet" refers to the family of networking technologies used for local area networks. (LANs)

- Ethernet is a set of protocols and hardware standards for connecting devices in a wired LAN.
- It is a technology used to move packets from one machine to another.
- It defines how data is transmitted and how devices communicate over a physical medium like twisted pair or fiber optic cables.

* Preamble: The first field of the 802.3 frame contains 7 bytes (56 bits) of alternating 0s and 1s ~~that~~ alerts the receiving system to the comming frame and enables it to synchronize its input ~~byt~~ timing.
- The pattern provides only an alert and a timing pulse.
- The preamble is actually added at the physical layer and is not part of the frame.

* Start frame delimiter (SFD):
- The second field (10101011 / 0xAB) signals the beginning of the frame.
- The SFD warns station or stations that this is the last chance for synchronization.
- The last 2 bits is 11 and alerts the receiver that the next field is the destination address.

* Destination Address:
The destination address field is 6 bytes and contains the physical address of the destination station or stations to receive the packet. (MAC Address of recipient)

* Source Address:
The SA field is also 6 bytes and contains the physical address (mac) of the sender of the packet.

* length or type:
- This field is defined as a type field or length field.
- If the value is ≤ 1500 (0x05DC):
    - It is interpreted as length of the payload.
    - This type of frame is called IEEE 802.3 Ethernet frame.

- If the value is ≥ 1536 (0x0600):
    - It is interpreted as Ether Type.
    - This tells the protocol used in the payload.
    - This is the Ethernet II frame format.
- Values between 1501 and 1535 are undefined.

0x0800	IPv4
0x0806	ARP
0x86DD	IPv6.
0x8035	RARP

* Data :-
  - This field carries data encapsulated from the upper-layer protocols
  - It is a minimun of 46 and maximum of 1500 bytes.

* CRC.
  - The last field contains error detection information.

* Addressing:
- Each station on an ethernet network has its own network interface card (NIC)
- The NIC provides 6-byte physical address. (48 bits) written in hexadecimal notation with a colon between the bytes.

ex. 06 : 01 : 02 : 01 : 2C : 4B.

- A source address is always a unicast address (the frame comes from only one station.)
- The destination address can be unicast, multicast, or broadcast.

- If the least significant bit (lsb) of the first byte in a destination address is 0, the address is unicast, otherwise, it is multicast

$\longrightarrow$ 0 $\longrightarrow$ Unicast

1 $\longrightarrow$ Multicast

- A unicast destination address defines only one recipient, the relationship between the Sender and the receiver is one to one.
- A multicast destination address defines a group of addresses, the relation between sender and receivers is one-to-many.

- The broadcast address is a special case of the multicast address, the recepients are all the stations on the LAN.

- A broadcast destination address is 48 1s i.e all Fs.

FF : FF : FF : FF : FF : FF.

★ Medium Access Control:

- It is a Sublayer of the Data Link layer defined by the IEEE 802 project
- It defines the access method and access control in different local area network protocols.

LLC
MAC

Data Link layer.

- Its main duty is to control access to a network's shared communication medium, ensuring that various devices can transmit data effectively and fairly over a common communication path like a wired Ethernet or wireless Wi-Fi network.

Medium Access Control

Controlled Access
- Reservation
- Polling
- Token passing.

Channelization
- FDMA
- TDMA
- CDMA

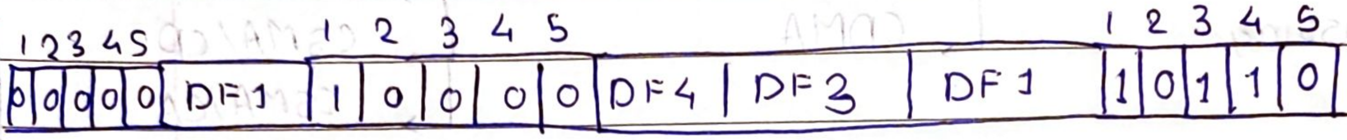
Random access
- ALOHA
- CSMA
- CSMA/CD
- CSMA/CA

* Controlled Access
- In controlled access, the stations consut one another to find which station has the right to send.
- A station cannot send unles it has been authorized by other stations.

1. Reservation:
- In the reservation method, a station needs to make a reservation before sending data.
- Time is divided into intervals. In each interval, a reservation frame precedes the data frames sent in that interval.
- If there are N stations in the system, there are exactly N reservation minislots in the reservation frame. Each minislot belongs to a station.
- When a station needs to send a data frame it makes reservation in its own minislot.
- The station that has made reservations can send their data frames after the reservation frame.

Ex. Below is a Situation with 5 Stations.

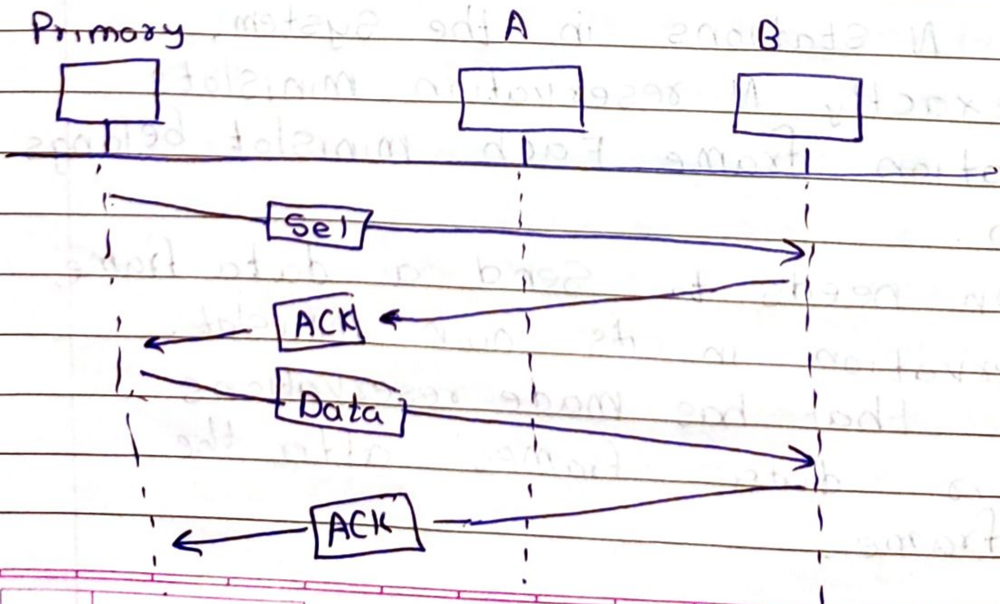
- In the first interval, only Station 1,3,4 have made reservation.
- In the second interval, only Station 1 has made a reservation



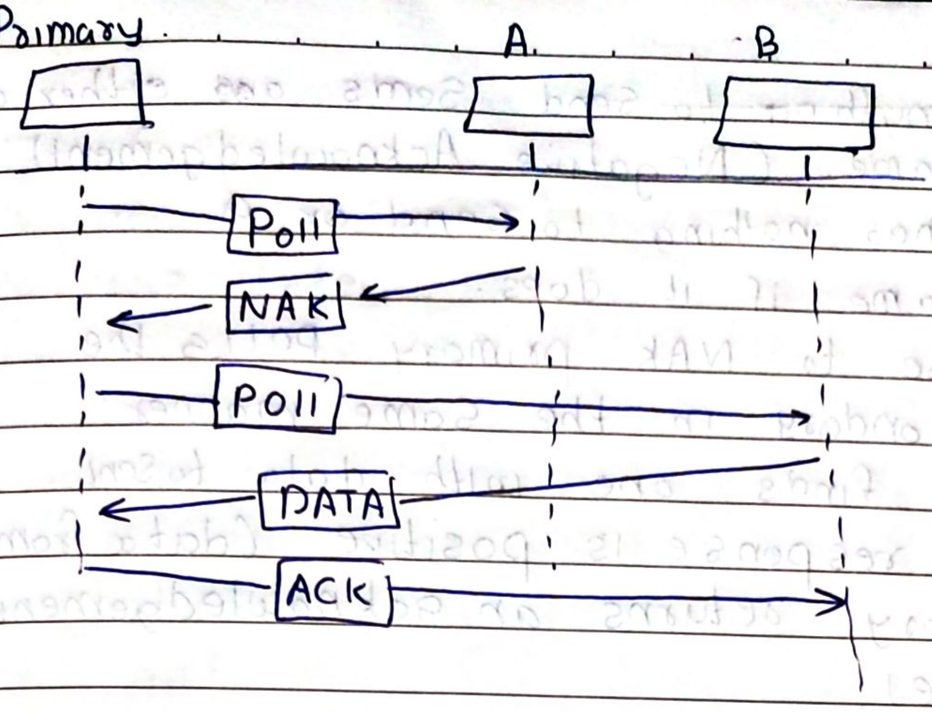
* Polling

- Polling works with topologies in which one device is designated as a primary station and other devises are secondary stations.
- All data exchanges must be made through the primary device even when the ultimate ~~desy~~ destination is a secondary device.

- The primary device controls the link. It is up to the primary device to determine which device is allowed to use the channel at a given time.
- The primary device is therefore always the initiator of a Session.



Select



POLL

* Select:
- The select function is used whenever the primary device has something to send.
- Before sending data, the primary creates and transmits a select (SEL) frame, one field of which includes the address of intended secondary.
- If the intended secondary is ready it returns an acknowledgement.
- After receiving to ack signal the primary station sends the data frame again on receiving of which an ACK frame is returned.

* POll
- The poll function is used by the primary device to get data from the secondary devices.
- When the primary is ready to receive data, it must ask (pool) each device int turn

~~if it has anything to send~~ sends ~~and~~ either a NAK frame (Negative Acknowledgement) if it has nothing to send or a data frame if it does.

- In response to NAK primory polls the next secondary in the same manner until it finds one with data to send.
- When the response is positive (data frame), the primary returns an acknowledgement (ACK frame).

★. Token Passing :

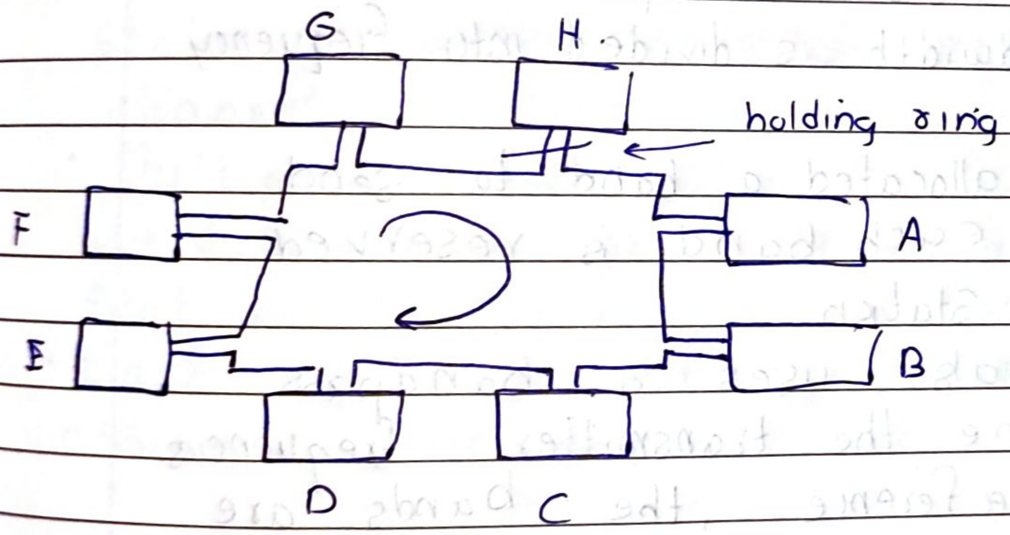
- In the token-passing method, the stations in a network are organized in a logical ring.
- In other words, for each station, there is a predecessor and a successor.
- The predecessor is the station which is logically before the station in the ring similarly successor is station which is after.

- In this method, a special packet called a token ring frame is circulated through the ring.
- The possession of the token gives the station the right to access the channel and set its data.

- When a station has some data to send, it waits until it receives the token ring from its predecessor.
- It then holds the token and sends its data.

- When the station has no more data to send, it releases the token, passing it to its successor.
- Holding the token makes a station gain control over the network.

- When a device receives a data frame frame from its predecessor, it checks the destination address.
- If the data is for some other device it passes it to its successor or else it receives the data if dest Address is its own.



Dest Add	Source Add	Data

Data frame

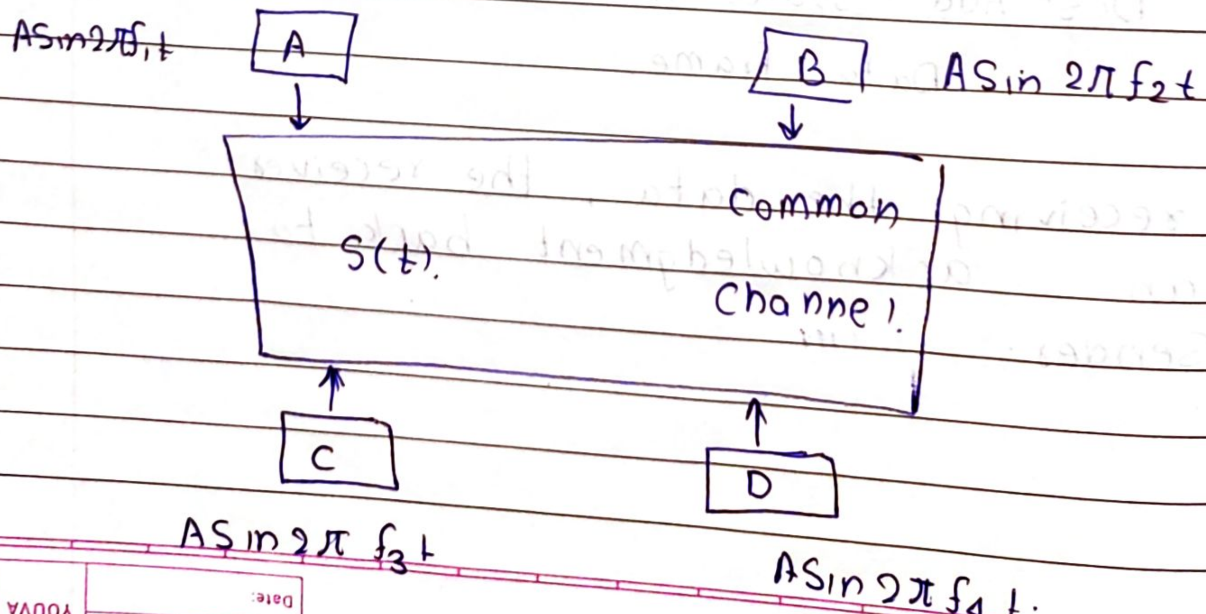
- After receiving the data, the receiver sends an acknowledgment back to the sender.

* Channelization :

- In this, the available bandwidth of the link is shared in time, frequency and code to multiple stations to access channel simultaneosly.
- It allows multiple users or devices to share the same communication medium (channel) efficiently.

1. Frequency - Division Multiple Access (FDMA)

- In frequency - division multiple access, the available bandwidth is divided into frequency bands.
- Each station is allocated a band to send its data. (i.e each band is reserved for a specific station.
- Each station also uses a bandpass filter to confine the transmitter frequencies.
- To prevent interference, the bands are seperated by unused spaces called guard bands.

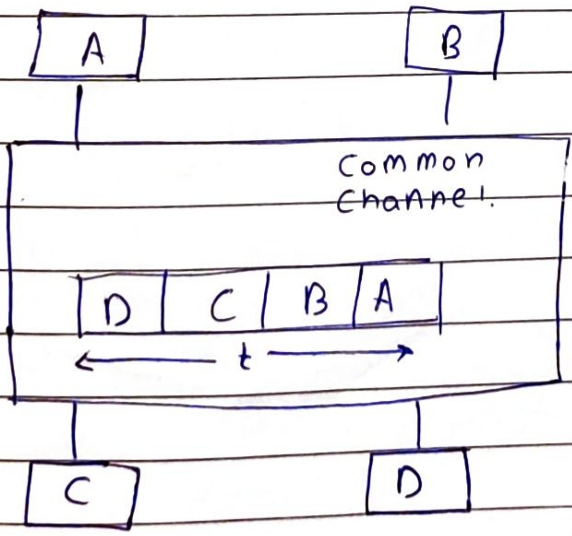


Combined Signal.

$$s(t) = A\sin 2\pi f_1 t + A\sin 2\pi f_2 t + A\sin 2\pi f_3 t + A\sin 2\pi f_4 t.$$

The receivers use filters and demodulation to extract its data.
ex. filter of f_1 only ~~receives~~ allows f_1 and block others.

2. Time Division Multiple Access (TDMA)
- In time division multiple access the stations share the bandwidth of the channel in time
- Multiple Stations share same frequency band, but each gets a specific time slot to transmit.
- Each Station transmits its data in its assigned time slot.



- It requires Synchronization between different Stations.

- Each station needs to know the beginning of its slot and the location of its slot.
- It is difficult because of propagation delays.

3. Code Division Multiple Access (CDMA)
- It uses only one channel which occupies the entire bandwidth of the link.
- All stations can send data simultaneously, there is not time sharing
- Here each user is allocated an unique code
- Orthogonal code is used.
  ie if C_1 is orthogonal
  $C_1 \cdot C_1$ = No of stations.

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